

Mobile Application Development with Flutter

Lecture 20: InkWell & GestureDetector

Introduction

- In many mobile applications, we want widgets such as **containers, images, or text** to respond when a user taps them
- Flutter provides a widget called **InkWell** that allows us to make any widget **clickable**
- **InkWell** also provides a **ripple effect animation** when the user taps the widget
- This ripple animation follows **Material Design guidelines**

What is InkWell?

- InkWell is a **gesture widget** that detects user interaction such as:
 - Tap
 - Double tap
 - Long press
- and displays a **ripple (splash) effect**.
- Example ripple effect:
 - User Tap → Ripple animation → Action performed

Basic Syntax

```
InkWell(  
  onTap: () {},  
  child: Widget(),  
)
```

- Explanation:

Property	Purpose
onTap	action when user taps
child	widget that becomes clickable

Example

- Demonstrate basic example
- InkWell with different events
- Splash color
- Highlight color
- InkWell with border radius

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GestureDetector Widget in Flutter

- In Flutter applications, we often need to detect **user gestures** such as:
 - Tap, Double tap, Long press, Drag, Swipe
- Flutter provides a widget called **GestureDetector** that allows us to detect these gestures on any widget
- Unlike InkWell, **GestureDetector** does not provide any **visual animation**
- It only detects gestures and triggers actions

Basic Syntax

```
GestureDetector(  
  onTap: () {},  
  child: Widget(),  
)
```

- Explanation:

Property	Purpose
onTap	called when user taps
child	widget that receives gesture

Example

- Demonstrate basic example
- GestureDetector with different events
- Detecting drag gesture
 - Horizontal drag
 - Vertical drag
 - Pan gesture
 - Swipe detection

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GestureDetector vs InkWell

Feature	InkWell	GestureDetector
Gesture detection	Yes	Yes
Ripple animation	Yes	No
Material design effect	Yes	No
Used for UI buttons	Yes	Usually No
Used for custom gestures	Limited	Yes

When to Use InkWell & GestureDetector

- Use InkWell when:
 - You want ripple animation
 - Following Material Design
 - Creating buttons or clickable cards
- Use GestureDetector when:
 - You need custom gesture detection
 - Detect drag or swipe
 - You don't need ripple animation